



**UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION**

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450/ITS432
EXAMINATION	:	JULY 2025
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of five (5) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any materials into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
 - i) the Question Paper
 - ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

QUESTION 1

- a) Data is utilised to create information, which serves as the foundation for effective decision-making. Database applications are developed to streamline the process of accessing, managing, and interacting with data. Identify **FOUR (4)** examples of database applications. (4 marks)
- b) There are several methods to determine the classification of the database. One of the methods is based on location. It can determine whether the database is classified as centralised database or distributed database. Briefly explain **TWO (2)** differences between centralised database and distributed database. (4 marks)
- c) Several different types of keys are used in the relational table. Briefly describe the candidate key, primary key, and secondary key accordingly. Then, give an example for each of the keys based on the data given in Table 1.

Table 1 : Agent Table

AgentID	Agent FirstName	Agent LastName	Passport Number	AreaCode	PhoneNumber
503	Alex	Alby	A12345678	615	894-1238
504	Myron	Orlando	B98765432	713	442-3391
505	Alex	John	C12398765	615	844-3809
506	Anne	Farriss	D45678901	713	443-2180

(6 marks)

- d) The selection of DBMS software is critical to the information system's smooth operation. Briefly explain **THREE (3)** common factors that are important in DBMS software selection. (6 marks)

QUESTION 2

- a) The network model was created to represent complex data relationships more effectively than the hierarchical model. However, as information needs grew and more sophisticated databases and applications were required, the network model became too cumbersome. Describe **TWO (2)** disadvantages of network model. (4 marks)

- b) Table 2 shows the data from a publishing company. Briefly explain **TWO (2)** reasons why this table violates the characteristics of a relational table.

Table 2: Books table

ISBN	BookTitle	Authors	Publisher ID	Publisher Name
978-981-4824-24-7	Database Systems	Coronel, Morris	101	Cencage Learning
978-967-388-341-7	Innovation via Arduino	Hafizah Hussain	103	John Wiley & Sons
978-967-86-1017-9	Healing the Invisible Wound	Ahmad Rostam, Rozaniza, Alizi Alias	102	Karangkraf Media Group
978-981-4824-24-7	Digital Multimedia	Nigel, Jenny	102	Karangkraf Media Group

(4 marks)

- c) In a database design process, there are two views of data within the system, which are the business view and the designer's view. Briefly explain **THREE (3)** views for the designer.

(6 marks)

- d) Describe **THREE (3)** benefits of big data to companies.

(6 marks)

QUESTION 3

Company IT Creative Solution makes an initiative to record its employees' skills. Table 3 shows a sample record for some of its employees. The company will request up to 2 important skills from each employee. However, this record is still in poor design. Some employees have not passed a certification test in any skill, while others have been certified in several skills. Some certified skills are shared by several employees, while other skills have no employees that hold those certifications.

Table 3 : Employee Skills Records of IT Creative Solution in 2025

Item	Staff ID	Staff Name	Position	Department ID & Name	Skill ID & Skill Name	Completion Date
1.	304	Intan Syafinaz	Analyst	D1 - Data Analytic	AD - Advanced Database	1/1/2025
2.	304	Intan Syafinaz	Analyst	D1 - Data Analytic	BM - Basic DB Manipulation	3/1/2025
3.	301	Robert James	UI/UX Designer	D2 - Software development	AD - Advanced Database	1/1/2025
4.	302	Mohd Fikri Mohd Ali	Developer	D2 - Software development	-	-
5.	307	Zinnirah	Developer	D2 - Software development	AD - Advanced Database	1/1/2025
6.	307	Zinnirah	Developer	D2 - Software development	FF - Frontend Frameworks	Not complete yet
7.	305	Linda Clark	System Admin	D3 - Network and Systems Administration	BM - Basic DB Manipulation	3/1/2025

Based on the information given, answer the following questions:

- State the normal form of Table 3. Explain your answer with reasons. (3 marks)
- Identify the primary key(s) and partial dependencies and/or transitive dependencies. (6 marks)
- Normalize Table 1 from first normal form (1NF) until third normal form (3NF). Show all the steps involved. (11 marks)

QUESTION 4

Table 4 shows a relational schema for XY Company, a consulting company that tracks all assigned projects.

Table 4 : Relational Schema for XY Company

EMPLOYEE (EmpID, EmpFirstName, EmpLastName, EmpHireDate, JobCode)

JOB (JobCode, JobName, JobChargeHour)

PROJECT (ProjCode, ProjName)

ASSIGN (AssignID, AssignDate, AssignHours, ProjCode, EmpID)

Write the SQL statements based on the following requirements:

- Display the project names that were assigned from January 2025 until April 2025.
(4 marks)
- Display the total charge for each employee and name the output as "Total Charge".
(5 marks)
- Display the full names of the employees who have been assigned to more than two projects.
(5 marks)
- By using a subquery, display the job name that has a charge per hour of RM50 and has more than 10 employees.
(6 marks)

QUESTION 5

Clean Environment Sdn. Bhd is a new company located in Jasin, Malacca. This company is planning to develop a solid waste recycling database system for better coordination and reporting on the monitoring, collection, and disposal of waste from each locality. As a database designer, you are tasked to design the database system according to the following requirements:

- Record the details of the collector, such as their collector ID, name, and phone number.
- Among the collectors, several are supervisors. Each collector is managed by one supervisor, and each supervisor manages many collectors.
- Each collector is assigned to several collection duties. Each duty involves only one collector. The date for the collection is recorded in the system.

- Each collection duty session is for one locality and uses one vehicle.
- Locality details, such as locality ID, locality name, and postcode, are recorded in the system.
- Vehicle details, such as the plate number, model, and size, are recorded in the system.
- Each locality contains several houses, and each house is located in one locality. The address and category of each house are recorded in the system.
- Each collection duty session collects various types of solid waste.
- Each piece of solid waste has an ID, name, and category.
- The total weight of the collected solid waste is also recorded in the system.

Based on the given information:

- a) Construct a complete Entity Relationship Diagram (ERD) with all of the required tables, attributes, and relationships using the Crow's Foot notation that reflects all of the given business rules. (16 marks)
- b) Identify **TWO (2)** relevant information that you can get from the ERD you constructed in (a). (4 marks)

END OF QUESTION PAPER